

Telekinetic Drive: Controlling Robots with Intent-Leaking Micro-Motions

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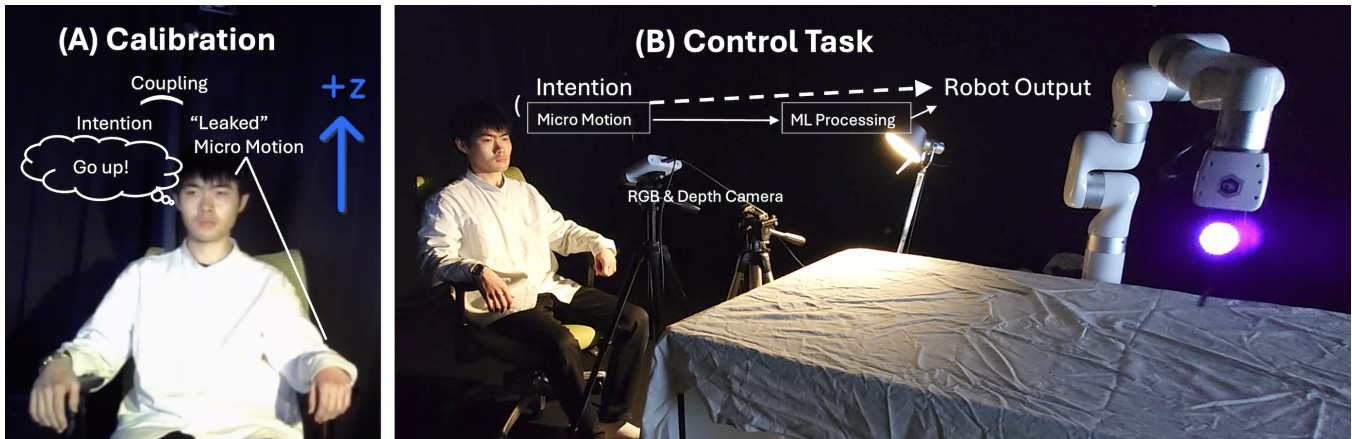


Figure 1: System overview. (A) Through a calibration process, the system learns the mapping between participants' micro-motions leaking during intention and the robots' movements. (B) Micro-motions associated with the participants' intentions are tracked, causing the robot arm to move in the corresponding directions. As a result, the participant's intention is directly coupled to the robot's motion.

Abstract

We propose Telekinetic Drive, a robot control system that directly connects user intent with robot output by leveraging subtle bodily micro-motions that unconsciously emerge when a user intends to control the robot. The system extracts features from micro-motions of the face and other body parts that occur when the user intends the robot to move in a specific direction, and learns the correspondence between these features and the robot's movement directions through machine learning.

The system is constructed using only camera based measurement and image analysis, and does not employ brain signals, gaze tracking, or electromyography for intent inference. Participants can operate a robot arm through mental effort alone, without wearing

any devices or intentionally moving their bodies for control. This enables an experience akin to controlling the robot by telekinesis.

CCS Concepts

• Human-centered computing → Interaction paradigms; Interaction techniques.

Keywords

robot control, intent inference, micro-motions, human-robot interaction, machine learning

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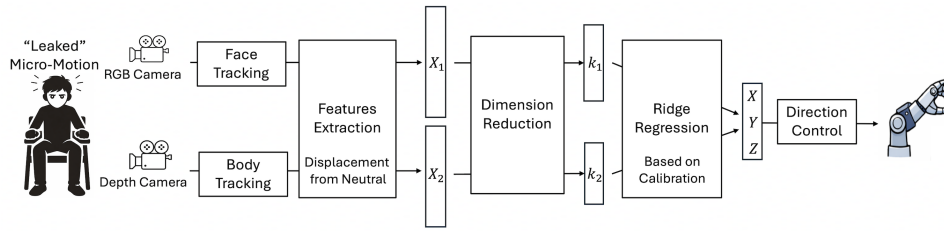


Figure 2: System Architecture of Telekinetic Drive.

1 Introduction

Over the past several decades, human–computer interaction (HCI) has been shifting toward forms that do not require explicit physical movements as input. One of the most prominent examples is the brain–computer interface (BCI), which offers potential advantages such as reducing physical fatigue and enabling interaction with lower physical burden [Maiseli et al. 2023; Wolpaw et al. 2002]. However, stable signal acquisition in BCI systems still requires large-scale equipment and complex preprocessing, making everyday use difficult [Wang et al. 2023].

On the other hand, systems that utilize peripheral bodily movements, such as facial expressions [Fukuoka et al. 2019], finger force [Hirao and Hashimoto 2025], eye gaze [Shafti et al. 2019], and shoulder movements [Shimobayashi et al. 2021], have demonstrated that machine learning can be used to provide an operational experience that feels easy to control for each individual. Nevertheless, these approaches require explicit bodily movements to mediate the intended control, and thus the connection between intent and control output is not direct. Therefore, there remains a need for systems that more directly link intent to control without requiring specialized equipment.

To address this challenge, we propose Telekinetic Drive. This approach senses bodily information that humans unconsciously leak when intending to control an object, without requiring brain activity measurement or consciously performed physical movements as a means of operation. By learning the coupling between intention and micro-motion through machine learning, the system enables control of a robotic arm. Telekinetic Drive leverages individual-specific patterns inherent in the manifestation of intent as subtle bodily micro-motions, enabling machine operation that adapts to a user’s physical characteristics and habits.

2 Implementation

Our system enables robotic arm control by learning the coupling between user intention and micro-motions (Figure 2).

During calibration, the system learns the mapping between intended motion directions and bodily micro-motions. For each of the six directions ($\pm x$, $\pm y$, $\pm z$), facial and bodily micro-motions are captured using cameras. Facial micro-motions are processed by extracting facial feature points and analyzing their planar displacements, while micro-motions of other body parts are analyzed in terms of joint-level displacement and tilt. The extracted features are reduced in dimensionality using Principal Component Analysis (PCA), combined, and learned using Ridge regression.

During operation, the captured micro-motions are regressed using the learned model to estimate the intended motion direction. The robotic arm is then assigned a velocity proportional to the magnitude of the regression output, and its end-effector position is updated in real time.

3 User Experience

Participants sit in a chair with their bodies immobilized and simply intend for the robotic arm to move. They first undergo a calibration phase, where the system learns the relationship between their bodily micro-motions and the intended movement directions of the robotic arm. Based on camera-captured micro-motions, the system infers the intended direction of movement and actuates the robot arm accordingly. The mapping is adjusted by recalibration as needed to better match each participant’s operability. Since no devices need to be worn and no deliberate bodily movements are required for control, participants experience the robot arm moving as if by telekinesis, or they can also experience the training process to improve their way of control. In the demonstration, participants use the system to manipulate objects.

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